

Aggregate Bounds on the Eigenvalues of Principal Submatrices and Majorization Relations

Professor Hristo Sendov **Mengxu Yuan**

Department of Mathematics
Western University
London, Ontario, Canada

January 25, 2026

This presentation is based on the following paper:

Aggregate bounds on the eigenvalues of the principal submatrices of a Hermitian matrix and majorization relations

Hristo Sendov & Mengxu Yuan

DOI / Link:

<http://arxiv.org/abs/2601.16320>

LE MENU MATHÉMATIQUE

An intellectual gastronomic experience

1 Apéritif

2 Entrée

3 Plat Principal

4 Fromage

5 Dessert

6 References

Bon appétit mathématique!

Apéritif

Majorization Theory

The concept of **Majorization** is central in matrix analysis.

Definition (Majorization, $\succ$)

For $x, y \in \mathbb{R}^N$, let $x^\downarrow$ and $y^\downarrow$ be sorted in non-increasing order. We say x **majorizes** y , denoted $x \succ y$, if:

$$\sum_{i=1}^k x_i^\downarrow \geq \sum_{i=1}^k y_i^\downarrow \quad (k < N) \quad \text{and} \quad \sum_{i=1}^N x_i = \sum_{i=1}^N y_i.$$

Example

Let $\mathbf{x} = (5, 3, 1)$ and $\mathbf{y} = (4, 3, 2)$. Both sum to 9.

- Partial sums of x : **5**, $5 + 3 = \mathbf{8}$, $5 + 3 + 1 = \mathbf{9}$.
- Partial sums of y : **4**, $4 + 3 = \mathbf{7}$, $4 + 3 + 2 = \mathbf{9}$.

Checking inequalities: $\mathbf{5} \geq \mathbf{4}$ and $\mathbf{8} \geq \mathbf{7}$. Thus, $\mathbf{x} \succ \mathbf{y}$.

Example

For any normalized vector $x \in \mathbb{R}_+^n$ (where $\sum x_i = 1$):

$$(1, 0, \dots, 0) \succ x \succ \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)$$

How to visualize $x \succ y$?

Think of y as a "**smoothed**" or "**averaged**" version of x .

- **The "Robin Hood" Operation:** If we transfer a small amount δ from a *larger* component x_i to a *smaller* component x_j (making them closer), the new vector moves *down* the order:

$$x_{\text{new}} \prec x_{\text{old}}$$

- **Entropy Connection:** Majorization captures the idea of information loss.

$$x \succ y \implies \text{ShannonEntropy}(x) \leq \text{ShannonEntropy}(y)$$

Majorization is a **Preorder** on $\mathbb{R}^n$, because it satisfies:

① **Reflexivity:** $x \succ x$.

② **Transitivity:** $x \succ y$ and $y \succ z \implies x \succ z$.

The Missing Piece: Antisymmetry

A partial order requires: $x \succ y$ and $y \succ x \implies x = y$.

Majorization fails this because it is **permutation invariant**.

Example

Let $x = (1, 0)$ and $y = (0, 1)$.

$$x^\downarrow = (1, 0) = y^\downarrow \implies x \succ y \text{ and } y \succ x$$

But clearly $x \neq y$. They are distinct, yet equivalent under $\succ$.

Note: If we restrict the domain to the cone of **sorted vectors**
 $\mathcal{D}_\downarrow = \{x \in \mathbb{R}^n : x_1 \geq \dots \geq x_n\}$, then $\succ$ becomes a **Partial Order**.

Intuition: Majorization moves from a "spread out" vector x to a more "averaged" vector y .

Schur's Theorem

For any $n \times n$ Hermitian matrix A , the vector of eigenvalues $\lambda(A)$ majorizes the vector of diagonal entries $\text{diag}(A)$:

$$\lambda(A) \succ \text{diag}(A).$$

Insight: Since diagonal entries are just 1×1 principal submatrices, Schur's theorem relates the spectrum of A (size n) to its submatrices of size 1.

Majorization controls not just sums, but also **products** of non-negative vectors.

Proposition (Schur-Concavity of the Product)

If $x, y \in \mathbb{R}_+^n$ (non-negative vectors) and $x \succ y$, then:

$$\prod_{i=1}^n x_i \leq \prod_{i=1}^n y_i$$

Example (Consequence: Hadamard's Inequality)

Let A be positive semidefinite. By Schur's Theorem, $\lambda(A) \succ \text{diag}(A)$. Applying the product inequality:

$$\det(A) = \prod_{i=1}^n \lambda_i(A) \leq \prod_{i=1}^n A_{ii}$$

This recovers the classical **Hadamard's Inequality**.

Apéritif: Eigenvalues of Principal Submatrices

We aim to generalize relationships between A and its submatrices of *any* size. Let A be an $n \times n$ Hermitian matrix.

- **Eigenvalues of A :** $\lambda(A) = (\lambda_1, \dots, \lambda_n)$ with $\lambda_1 \geq \dots \geq \lambda_n$.
- **Submatrix A_k :** The $(n-1) \times (n-1)$ principal submatrix obtained by deleting the k -th row and column.
- **Eigenvalues of A_k :** $\mu(A_k) = \lambda(A_k) = (\mu_{k,1}, \dots, \mu_{k,n-1})$.

Cauchy's Interlacing Theorem

The eigenvalues of A and *any individual* principal submatrix A_k satisfy:

$$\lambda_1 \geq \mu_{k,1} \geq \lambda_2 \geq \mu_{k,2} \geq \dots \geq \mu_{k,n-1} \geq \lambda_n$$

Cauchy's Interlacing Theorem describes one submatrix at a time. What about the **aggregate behavior** of all submatrices $A_1, \dots, A_n$?

Theorem (Thompson, 1966)

For any fixed index j ($1 \leq j \leq n-1$), the sum of the j -th eigenvalues satisfies:

$$\lambda_j + (n-1)\lambda_{j+1} \leq \sum_{k=1}^n \mu_{k,j} \leq (n-1)\lambda_j + \lambda_{j+1}.$$

The Research Question: Can we extend Thompson's bounds to sums over a **range of indices** $[\ell, r]$?

$$\sum_{k=1}^n \sum_{j=\ell}^r \mu_{k,j} \quad \text{vs.} \quad \text{Sums of } \lambda$$

Let's try the **Naive Approach**: simply summing Thompson's upper bound for $j = \ell, \dots, r$.

$$\text{Naive UB} = \sum_{j=\ell}^r [(n-1)\lambda_j + \lambda_{j+1}] = (n-1) \sum_{j=\ell}^r \lambda_j + \sum_{j=\ell}^r \lambda_{j+1}$$

Our New Aggregate Bound (to be proved):

$$\text{Our UB} = (n-1) \sum_{j=\ell}^r \lambda_j + (r - \ell + 1) \lambda_{r+1}$$

The Improvement Gap

Let's subtract Our Bound from the Naive Bound:

$$\text{Naive} - \text{Ours} = \sum_{j=\ell}^r \lambda_{j+1} - (r - \ell + 1) \lambda_{r+1} = \sum_{j=\ell}^r (\lambda_{j+1} - \lambda_{r+1}) \geq 0$$

*Conclusion: Since $\lambda_{j+1} \geq \lambda_{r+1}$, our bound is **strictly sharper** (smaller) than the naive sum!*

We derived the following **aggregate bounds** for the aggregate sum of eigenvalues over a window $[\ell, r]$.

Theorem (Theorem 1.4)

For any $1 \leq \ell \leq r \leq n-1$:

$$\underbrace{(r - \ell + 1)\lambda_\ell + (n - 1) \sum_{j=\ell}^r \lambda_{j+1}}_{\text{Lower Bound}} \leq \sum_{k=1}^n \sum_{j=\ell}^r \mu_{k,j}$$

$$\leq (n - 1) \sum_{j=\ell}^r \lambda_j + \underbrace{(r - \ell + 1)\lambda_{r+1}}_{\text{Upper Bound}}$$

This implies the following **relaxed** version:

Corollary (Corollary 1.5)

$$\begin{aligned}(n - \ell)\lambda_\ell + (n - 1) \sum_{j=\ell+1}^r \lambda_j + r\lambda_n \\ \leq \sum_{k=1}^n \sum_{j=\ell}^r \mu_{k,j} \\ \leq (n - \ell)\lambda_1 + (n - 1) \sum_{j=\ell+1}^r \lambda_j + r\lambda_{r+1}\end{aligned}$$

This corollary reproves a known result:

Theorem (Johnson & Robinson, 1981)

$$\max_{1 \leq i \leq n} \mu_{i,j} \geq \frac{n-j}{n} \lambda_j + \frac{j}{n} \lambda_n \text{ and } \min_{1 \leq i \leq n} \mu_{i,j} \leq \frac{n-j}{n} \lambda_1 + \frac{j}{n} \lambda_{j+1}$$

Apéritif: Main Results

In order to get **aggregate bounds**, we need the following theorem which is related to calculus.

Let $\lambda_1 \geq \cdots \geq \lambda_n$ be reals and $w_i \geq 0$ be weights ($\sum_{i=1}^n w_i = 1$). Let $\mu_1 \geq \cdots \geq \mu_{n-1}$ be roots of

$$p(x) = \sum_{i=1}^n w_i \prod_{\substack{j=1 \\ j \neq i}}^n (x - \lambda_j).$$

For any window $[\ell, r]$, we derived the following bounds:

Theorem (Main Theorem: Theorem 1.6)

$$\sum_{j=\ell}^r \lambda_{j+1} + \sum_{j=\ell}^r \frac{w_{j+1}}{L_{r+1}} (\lambda_\ell - \lambda_{j+1}) \leq \sum_{j=\ell}^r \mu_j \leq \sum_{j=\ell}^r \lambda_j - \sum_{j=\ell}^r \frac{w_j}{U_\ell} (\lambda_j - \lambda_{r+1})$$

where $U_\ell := \sum_{i=\ell}^n w_i$ and $L_{r+1} := \sum_{i=1}^{r+1} w_i$.

To describe another main result, we need a notation for the collection of **all** eigenvalues at a specific level m .

Definition (The Vector $X_m(A)$)

Let A be an $n \times n$ Hermitian matrix. For any $1 \leq m \leq n$, let

$$X_m(A)$$

denote the vector containing the eigenvalues of **all** $\binom{n}{m}$ principal submatrices of size $m \times m$.

Properties:

- **Dimension:** The vector $X_m(A)$ has length $m \cdot \binom{n}{m}$.
- **Ordering:** We assume the entries are sorted non-increasingly.
- **Sum:** The sum of entries of $X_m(A)$ is $\binom{n-1}{m-1} \operatorname{tr}(A)$.

Definition (The Vector $X_m(A)$)

Let A be an $n \times n$ Hermitian matrix. For any $1 \leq m \leq n$, let

$$X_m(A)$$

denote the vector containing the eigenvalues of **all** $\binom{n}{m}$ principal submatrices of size $m \times m$.

Example (Extreme Cases)

- **Case** $m = n$: Only 1 submatrix (A itself).
 $\implies X_n(A) = \lambda(A)$ (The spectrum of A).
- **Case** $m = 1$: Submatrices are the 1×1 diagonal entries.
 $\implies X_1(A) = \text{diag}(A)$ (The diagonal entries).

Using the **aggregate bounds**, we established a complete **Spectral Hierarchy** between eigenvalues of submatrices of *any* two sizes m and k .

Theorem (Theorem 1.7)

Let $X_m(A)$ be the vector of eigenvalues of all $m \times m$ principal submatrices. For any $1 \leq k \leq m \leq n$:

$$\underbrace{X_m(A) \cup \cdots \cup X_m(A)}_{\binom{m-1}{k-1} \text{ times}} \succ \underbrace{X_k(A) \cup \cdots \cup X_k(A)}_{\binom{n-k}{m-k} \text{ times}}$$

- This majorization generalizes Schur's Theorem ($m = n, k = 1$).
- This majorization implies Szasz's Inequalities by applying product.

Schur's Theorem

For any $n \times n$ Hermitian matrix A , the vector of eigenvalues $\lambda(A)$ majorizes the vector of diagonal entries $\text{diag}(A)$:

$$\lambda(A) \succ \text{diag}(A).$$

Szasz's Inequalities

$$P_m(A)^{1/\binom{n-1}{m-1}} \leq P_k(A)^{1/\binom{n-1}{k-1}}$$

where $P_m(A)$ denote the product of all $m \times m$ principal minors of A , $1 \leq m \leq n$. For example $P_n(A) = \det(A)$ and $P_1(A) = A_{1,1} \cdots A_{n,n}$, the product of the diagonal entries of A .

Entrée

The Polynomial Setup

Lemma 2.1 (The Setup)

Let $\lambda_1 \geq \cdots \geq \lambda_n$ be real numbers (called **poles**).

Let $\omega_1, \dots, \omega_n$ be non-negative weights such that $\sum_{i=1}^n \omega_i = 1$.

Consider the polynomial:

$$p(x) := \sum_{i=1}^n \omega_i \prod_{\substack{j=1 \\ j \neq i}}^n (x - \lambda_j). \quad (1)$$

Key Property (Interlacing):

The polynomial $p(x)$ has $n - 1$ real roots, denoted $\mu_1 \geq \cdots \geq \mu_{n-1}$, which **interlace** with the poles:

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \cdots \geq \mu_{n-1} \geq \lambda_n.$$

The Rational Function Representation

If λ_i 's are distinct and ω_i 's are nonnegative, dividing $p(x)$ by $\prod(x - \lambda_j)$ gives the equivalent equation:

$$f(x) := \sum_{i=1}^n \frac{\omega_i}{x - \lambda_i} = \frac{p(x)}{\prod_{j=1}^n (x - \lambda_j)}. \quad (2)$$

Note:

- The roots μ_k 's are the zeros of this rational function $f(x)$ if λ_i 's are distinct and ω_i are nonnegative.
- f is strictly decreasing on each interval $(\lambda_{i+1}, \lambda_i)$ with $f(\lambda_{i+1}+) = \infty$ and $f(\lambda_i-) = -\infty$. Thus, $f(x)$ vanishes exactly once in each such interval. These vanishing points are zeros of $p(x)$ and they strictly interlace with the poles.

The Main Result

Our goal is to prove the main theorem which is related to calculus.

Theorem (Theorem 2.2)

For any indices ℓ and r with $1 \leq \ell \leq r \leq n-1$, assuming $L_{r+1}, U_\ell \neq 0$:

$$\begin{aligned} \sum_{j=\ell}^r \lambda_{j+1} + \sum_{j=\ell}^r \frac{\omega_{j+1}}{L_{r+1}} (\lambda_\ell - \lambda_{j+1}) &\leq \sum_{j=\ell}^r \mu_j \\ &\leq \sum_{j=\ell}^r \lambda_j - \sum_{j=\ell}^r \frac{\omega_j}{U_\ell} (\lambda_j - \lambda_{r+1}) \end{aligned}$$

where:

$$U_\ell := \sum_{i=\ell}^n \omega_i \quad \text{and} \quad L_{r+1} := \sum_{i=1}^{r+1} \omega_i.$$

Proof Strategy: Four Simplifying Assumptions

Proving Theorem 2.2 directly is cumbersome. We proceed by imposing four assumptions to simplify the problem, then relaxing them.

1. **Distinct Poles:** $\lambda_1 > \lambda_2 > \cdots > \lambda_n$.
2. **Positive Weights:** $\omega_i > 0$ for all i .
3. **Rational Weights:** $\omega_i \in \mathbb{Q}$ for all i .
4. **Equal Weights:** $\omega_i = \frac{1}{n}$ for all i .

Justifying Assumptions 1, 2 & 3

Are we allowed to make these assumptions? Yes, due to **continuity**.

Assumptions 1 & 2 (λ Distinctness & ω Positivity):

- These ensure roots of p are precisely roots of f .
- Since roots are continuous functions of λ_i 's, results for distinct λ_i 's hold for non-distinct λ_i 's in the limit.
- Since roots are continuous functions of ω_i 's, results for positive ω_i 's hold for nonnegative ω_i 's in the limit.

Assumption 3 (ω Rationality):

- The rational numbers $\mathbb{Q}$ are dense in $\mathbb{R}$.
- Any real weight ω_i is the limit of a rational sequence $\omega_i^{(k)} \rightarrow \omega_i$.
- Since our target inequalities are non-strict ($\leq$), they are preserved under limits.

Justifying Assumption 4: The Clustering Trick

Assumption 4 ($\omega_i = 1/n$) seems restrictive. Why is it sufficient?

Suppose assumptions 1, 2, 3:

$$\omega_i = \frac{k_i}{N}, \quad \text{where } k_i \in \mathbb{Z}^+ \text{ and } N = \sum k_i.$$

The defining equation becomes:

$$f(x) = \sum_{i=1}^n \frac{k_i/N}{x - \lambda_i} = \frac{1}{N} \sum_{i=1}^n \underbrace{\left(\frac{1}{x - \lambda_i} + \cdots + \frac{1}{x - \lambda_i} \right)}_{k_i \text{ times}}.$$

Interpretation: We view this as a system with N **poles** (where pole λ_i is repeated k_i times) and **equal weights** $1/N$.

Justifying Assumption 4: The Clustering Trick

The defining equation becomes:

$$f(x) = \sum_{i=1}^n \frac{k_i/N}{x - \lambda_i} = \frac{1}{N} \sum_{i=1}^n \underbrace{\left(\frac{1}{x - \lambda_i} + \cdots + \frac{1}{x - \lambda_i} \right)}_{k_i \text{ times}} = \frac{1}{N} \sum_{i=1}^n \sum_{j=1}^{k_i} \left(\frac{1}{x - \lambda_{ij}} \right)$$

where $\lambda_{ij} = \lambda_i$ for $j = 1, 2, \dots, k_i$.

However, this **violates** assumption 1 (λ distinct)!

So we replace λ_{ij} by $\lambda_{ij}(\epsilon) = \lambda_i + \frac{\epsilon}{j}$.

The Geometry of Equal Weights

When we assume $\omega_i = 1/N$ (Equal Weights), the structure becomes elegant.

Let $q(x) = \prod_{j=1}^N (x - \lambda_j)$ be the polynomial with roots at the poles. The polynomial $p(x)$ becomes:

$$p(x) = \frac{1}{N} \sum_{i=1}^N \prod_{j \neq i} (x - \lambda_j) = \frac{1}{N} q'(x).$$

Conclusion: Under Assumptions 1, 2, 3, 4, the roots μ are simply the **critical points** (zeros of the derivative) of the polynomial $q(x)$.

We first prove the bounds for this specific case, then map it back to the general case using the clustering ratio k_i/N .

Proof (Equal Weights): Step 1 - Monotonicity

We now prove the bounds assuming $\omega_i = 1/n$.

First, we show that roots μ move in the same direction as poles λ .

Differentiating $f(\mu_k) = \sum_{i=1}^n \frac{1}{\mu_k - \lambda_i} = 0$ with respect to λ_j gives:

$$\frac{\partial \mu_k}{\partial \lambda_j} = \frac{(\mu_k - \lambda_j)^{-2}}{\sum_{i=1}^n (\mu_k - \lambda_i)^{-2}} > 0.$$

Monotonicity Principle

Every root μ_k is a strictly increasing function of every pole λ_j .

Proof (Equal Weights): Step 2 - The Upper Bound

Since μ increases with λ , to maximize $\sum_{\ell}^r \mu_j$, we push the "outer" poles to their maximum valid positions:

- **Left:** Push $\lambda_1, \dots, \lambda_{\ell-1} \rightarrow +\infty$.
- **Right:** Push $\lambda_{r+2}, \dots, \lambda_n$ up to λ_{r+1} .

The roots $\mu_{\ell}, \dots, \mu_r$ become the roots of the derivative of:

$$\tilde{q}(x) = (x - \lambda_{r+1})^{n-r} \prod_{j=\ell}^r (x - \lambda_j).$$

Using Vieta's formulas on $\tilde{q}'(x)$, we obtain the explicit bound:

$$\sum_{j=\ell}^r \mu_j \leq \sum_{j=\ell}^r \lambda_j - \frac{1}{n - \ell + 1} \sum_{j=\ell}^r (\lambda_j - \lambda_{r+1}).$$

Conclusion: Recovering Theorem 2.2

Finally, we map the Equal Weight result back to General Weights using the clustering trick.

In the rational approximation ($\omega_j = k_j/N$), the coefficient from the Equal Weight bound was:

$$\frac{1}{\text{Count of poles from } \ell \text{ to } n}.$$

With clusters, the count of poles is $\sum_{i=\ell}^n k_i = NU_\ell$. However, the pole λ_j is repeated k_j times, so we sum the term k_j times:

$$\text{Total Coeff} = k_j \times \frac{1}{NU_\ell} = \frac{k_j/N}{U_\ell} = \frac{w_j}{U_\ell}.$$

This matches exactly the term in Theorem 2.2. □

This page is not very rigorous, please refer to section 2 of our paper.

The Main Result

The derivation of lower bound is similar. Thus we proved:

Theorem (Theorem 2.2)

For any indices ℓ and r with $1 \leq \ell \leq r \leq n-1$, assuming $L_{r+1}, U_\ell \neq 0$:

$$\begin{aligned} \sum_{j=\ell}^r \lambda_{j+1} + \sum_{j=\ell}^r \frac{\omega_{j+1}}{L_{r+1}} (\lambda_\ell - \lambda_{j+1}) &\leq \sum_{j=\ell}^r \mu_j \\ &\leq \sum_{j=\ell}^r \lambda_j - \sum_{j=\ell}^r \frac{\omega_j}{U_\ell} (\lambda_j - \lambda_{r+1}) \end{aligned}$$

where:

$$U_\ell := \sum_{i=\ell}^n \omega_i \quad \text{and} \quad L_{r+1} := \sum_{i=1}^{r+1} \omega_i.$$

Plat Principal

Matrix Setup

1. Spectral Decomposition

Let A be an $n \times n$ Hermitian matrix: $A = V\Lambda V^* = \sum_{i=1}^n \lambda_i v_i v_i^*$

- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ contains eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$.
- $V = [v_1, \dots, v_n]$ is unitary; v_i are the orthonormal eigenvectors.

2. Principal Submatrices

Let A_k be the $(n-1) \times (n-1)$ matrix obtained by deleting the k -th row and column of A .

- We denote the eigenvalues of A_k by:

$$\mu(A_k) = (\mu_{k,1} \geq \dots \geq \mu_{k,n-1}).$$

The Bridge to Matrices

The connection between matrix eigenvalues and polynomials is given by the following Lemma.

Lemma (Inspired from Golub 1973)

Let B be the compression of A onto the subspace $u^\perp$ (where u is a unit vector). The eigenvalues of B are the roots of:

$$\sum_{i=1}^n |\langle u, v_i \rangle|^2 \prod_{\substack{j=1 \\ j \neq i}}^n (x - \lambda_j) = 0. \quad (3)$$

Key Identification (Mapping)

This is **exactly** our polynomial $p(x)$ from the Entrée!

- **Weights:** $\omega_i = |\langle u, v_i \rangle|^2$.
- **Submatrices:** For A_k , we choose $u = e_k$, so $\omega_i = |v_{i,k}|^2$.

The Translation Dictionary

- **Weights:** $w_i \longrightarrow |\langle u, v_i \rangle|^2$ (Squared projection of u onto eigenvectors).
- **Roots:** $\mu_j \longrightarrow$ Eigenvalues of the compressed matrix B .

Theorem (Theorem 3.2: Renormalized Geometric Bounds)

For any unit vector u , the eigenvalues μ_j of the compression B satisfy:

$$\begin{aligned} \sum_{j=\ell}^r \lambda_{j+1} + \sum_{j=\ell}^r \frac{|\langle u, v_{j+1} \rangle|^2}{L_{r+1}(u)} (\lambda_{\ell} - \lambda_{j+1}) &\leq \sum_{j=\ell}^r \mu_j \\ &\leq \sum_{j=\ell}^r \lambda_j - \sum_{j=\ell}^r \frac{|\langle u, v_j \rangle|^2}{U_{\ell}(u)} (\lambda_j - \lambda_{r+1}). \end{aligned}$$

where

$$U_{\ell}(u) := \sum_{i=\ell}^n |\langle u, v_i \rangle|^2 \quad \text{and} \quad L_{r+1}(u) := \sum_{i=1}^{r+1} |\langle u, v_i \rangle|^2.$$

The problem of bounding eigenvalues of principal submatrices corresponds to a specific choice of the projection vector: $u = e_k$.

Theorem (Theorem 3.3)

Fix an index k . The eigenvalues $\mu_{k,j}$ of the submatrix A_k satisfy:

$$\sum_{j=\ell}^r \lambda_{j+1} + \sum_{j=\ell}^r \frac{|v_{j+1,k}|^2}{L_{r+1}^k} (\lambda_\ell - \lambda_{j+1}) \leq \sum_{j=\ell}^r \mu_{k,j} \leq \sum_{j=\ell}^r \lambda_j - \sum_{j=\ell}^r \frac{|v_{j,k}|^2}{U_\ell^k} (\lambda_j - \lambda_{r+1}),$$

provided the renormalization factors are non-zero:

$$U_\ell^k := \sum_{i=\ell}^n |v_{i,k}|^2 \quad \text{and} \quad L_{r+1}^k := \sum_{i=1}^{r+1} |v_{i,k}|^2. \quad (4)$$

Significance: This formula explicitly links the spectrum of the submatrix A_k to the k -th **components** of the original eigenvectors v_i .

What happens if we sum the eigenvalues of **ALL** principal submatrices?
Summing our previous result over $k = 1, \dots, n$ yields the following bounds.

Theorem (Theorem 3.4: Aggregate Renormalized Bounds)

For any window $[\ell, r]$, the total sum satisfies:

$$\begin{aligned} n \sum_{j=\ell}^r \lambda_{j+1} + \sum_{j=\ell}^r (\lambda_\ell - \lambda_{j+1}) \Psi_{j+1}(r) &\leq \sum_{k=1}^n \sum_{j=\ell}^r \mu_{k,j} \\ &\leq n \sum_{j=\ell}^r \lambda_j - \sum_{j=\ell}^r (\lambda_j - \lambda_{r+1}) \Omega_j(\ell), \end{aligned}$$

where Ω and Ψ are the global renormalization factors:

$$\Omega_j(\ell) := \sum_{k=1}^n \frac{|v_{j,k}|^2}{U_\ell^k} \quad \text{and} \quad \Psi_j(r) := \sum_{k=1}^n \frac{|v_{j,k}|^2}{L_{r+1}^k}.$$

The previous bounds depend on eigenvectors. Can we get a simpler, coordinate-free result? Yes! Since denominators are ≤ 1 , we have $\Omega_j(\ell) \geq 1$ and $\Psi_j(r) \geq 1$.

Theorem (Theorem 3.5)

*Relaxing the factors yields bounds depending **only** on eigenvalues:*

$$(r - \ell + 1)\lambda_\ell + (n - 1) \sum_{j=\ell}^r \lambda_{j+1} \leq \sum_{k=1}^n \sum_{j=\ell}^r \mu_{k,j} \leq (n - 1) \sum_{j=\ell}^r \lambda_j + (r - \ell + 1)\lambda_{r+1}.$$

Connection to History

When $\ell = r = j$, this collapses exactly to Thompson's 1976 Bounds:

$$(n - 1)\lambda_{j+1} + \lambda_j \leq \sum_{k=1}^n \mu_{k,j} \leq (n - 1)\lambda_j + \lambda_{j+1}.$$

Fromage

The Spectral Hierarchy Setup

To describe the hierarchy, we need precise notation for the collections of eigenvalues.

Definition (Vectors $X_m(A)$ and $Y_{n,m}(A)$)

Let A be an $n \times n$ Hermitian matrix.

- Let $X_m(A)$ be the vector containing the eigenvalues of **ALL** $\binom{n}{m}$ principal submatrices of size $m \times m$.
- Let $Y_{n,m}(A)$ be the vector containing $\binom{n-1}{m-1}$ copies of the eigenvalues of A itself ($\lambda(A)$).

Example

- $X_n(A) = \lambda(A)$ (The full spectrum).
- $X_1(A) = \text{diag}(A)$ (The diagonal entries).

The Spectral Hierarchy Theorem

This is the "Strong Cheese" of our menu: a complete majorization ordering between submatrices of any two sizes.

Theorem (Theorem 4.1: Spectral Hierarchy)

For any integers $1 \leq k \leq m \leq n$, the eigenvalues of size m majorize the eigenvalues of size k (with appropriate repetition):

$$\underbrace{X_m(A) \cup \dots \cup X_m(A)}_{\binom{m-1}{k-1} \text{ times}} \succ \underbrace{X_k(A) \cup \dots \cup X_k(A)}_{\binom{n-k}{m-k} \text{ times}}. \quad (5)$$

Meaning: The spectrum becomes "more spread out" as the submatrix size k decreases.

Recovering Classics

Our hierarchy unifies several classical results.

1. Schur's Theorem ($m = n, k = 1$)

The relation simplifies to:

$$\lambda(A) \succ \text{diag}(A).$$

Let $P_m(A)$ denote the product of all $m \times m$ principal minors of A .

2. Szasz's Inequalities

Since products are Schur-concave, majorization implies inequalities for products of elements. Our theorem recovers the famous Szasz's Inequalities for principal minors:

$$P_m(A)^{1/\binom{n-1}{m-1}} \leq P_k(A)^{1/\binom{n-1}{k-1}}, \quad \text{for } k \leq m.$$

Let's prepare some results to prove Theorem 4.1: Spectral Hierarchy.

Lemma (Lemma 4.3)

For any $n \geq 2$ and any $n \times n$ Hermitian matrix A :

$$Y_{n,n-1}(A) \succ X_{n-1}(A),$$

namely

$$\underbrace{\lambda(A) \cup \dots \cup \lambda(A)}_{n-1 \text{ times}} \succ X_{n-1}(A).$$

We have

$$\begin{aligned} \sum_{i=1}^{(n-1)n} (X_{n-1})_i &= \sum_{k=1}^n \sum_{j=1}^{n-1} \mu_{k,j} = \sum_{k=1}^n \operatorname{tr}(A_k) = (n-1) \operatorname{tr}(A) \\ &= (n-1) \sum_{j=1}^{n-1} \lambda_j + (n-1) \lambda_n = \sum_{i=1}^{(n-1)n} (Y_{n,n-1})_i. \end{aligned}$$

Proof Mechanism :

- ① **Nodal Check** ($p = rn$): First, we verify the majorization inequality at integer multiples of n .

$$\sum_{i=1}^{rn} (X_{n-1})_i \leq \sum_{i=1}^{rn} (Y_{n,n-1})_i.$$

This follows from the upper bound in Theorem 3.5, applied with $\ell = 1$:

$$\sum_{i=1}^{rn} (X_{n-1})_i = \sum_{k=1}^n \sum_{j=1}^r \mu_{k,j} \leq (n-1) \sum_{j=1}^r \lambda_j + r\lambda_{r+1} = \sum_{i=1}^{rn} (Y_{n,n-1})_i.$$

- ② **Unimodality**: Between the nodes, the difference $\Delta_p = \sum_{i=1}^p (Y_{n,n-1})_i - \sum_{i=1}^p (X_{n-1})_i$ rises and falls (due to Interlacing). Since it is non-negative at the nodes, it remains non-negative everywhere.

The Inductive Step

We now establish the structural link between dimension n and $n - 1$.

Lemma (Lemma 4.4: The Inductive Link)

Let $n \geq 2$. Suppose that

$$Y_{n-1,m}(B) \succ X_m(B)$$

holds for all $(n - 1) \times (n - 1)$ Hermitian matrices B .

Then,

$$Y_{n,m}(A) \succ X_m(A)$$

holds for all $n \times n$ Hermitian matrices A .

Significance: This allows us to use induction on the matrix size n . Since the base case ($n = 2$) is trivial, if we prove this lemma, the theorem holds for all n .

Proof of Lemma 4.4 (Aggregation)

Step 1: Apply Hypothesis to Submatrices

Let $A_1, \dots, A_n$ be the principal submatrices of size $(n-1)$. By assumption, for each k :

$$Y_{n-1,m}(A_k) \succ X_m(A_k).$$

By the **scaling property** of majorization, the union over all k preserves the ordering:

$$\bigcup_{k=1}^n Y_{n-1,m}(A_k) \succ \bigcup_{k=1}^n X_m(A_k). \quad (6)$$

We must now analyze the composition of the LHS and RHS vectors counting the multiplicities.

Proof of Lemma 4.4 (Counting)

Step 2: Analyze the LHS

- Each term $Y_{n-1,m}(A_k)$ contains $\binom{n-2}{m-1}$ copies of $\lambda(A_k)$.
- The union $\bigcup_{k=1}^n \lambda(A_k)$ is exactly the vector $X_{n-1}(A)$.
- **Total LHS:** $\binom{n-2}{m-1}$ copies of $X_{n-1}(A)$.

Step 3: Analyze the RHS

- The union collects eigenvalues of $m \times m$ submatrices found within each parent A_k .
- An $m \times m$ matrix M is contained in A_k if the index deleted to form A_k is **not** in M .
- There are $n - m$ such available indices. Thus, every M is counted exactly $n - m$ times.
- **Total RHS:** $(n - m)$ copies of $X_m(A)$.

Proof of Lemma 4.4 (Conclusion)

Substituting these counts back into the majorization relation:

$$\underbrace{X_{n-1}(A) \cup \cdots \cup X_{n-1}(A)}_{\binom{n-2}{m-1} \text{ times}} \succ \underbrace{X_m(A) \cup \cdots \cup X_m(A)}_{n-m \text{ times}}.$$

Step 4: Transitivity

From Lemma 4.3, we know $\lambda(A)$ (repeated $n-1$ times) $\succ X_{n-1}(A)$.

Substituting this into the LHS:

$$\underbrace{\lambda(A) \cup \cdots \cup \lambda(A)}_{\binom{n-2}{m-1}(n-1) \text{ times}} \succ \underbrace{X_m(A) \cup \cdots \cup X_m(A)}_{n-m \text{ times}}.$$

Step 5: Cancellation

Divide the repetition counts by $(n-m)$. The new count for $\lambda(A)$ is:

$$\frac{n-1}{n-m} \binom{n-2}{m-1} = \frac{n-1}{n-m} \cdot \frac{(n-2)!}{(m-1)!(n-m-1)!} = \binom{n-1}{m-1}.$$

This matches the definition of $Y_{n,m}(A)$.

Theorem 4.5: General Majorization

We now state the general relationship between the eigenvalues of A and the submatrices of a fixed size m .

Theorem (Theorem 4.5)

For any $n \times n$ Hermitian matrix A and any submatrix size $1 \leq m \leq n$:

$$Y_{n,m}(A) \succ X_m(A).$$

Recap of Notation:

- $X_m(A)$: Vector of eigenvalues of all $\binom{n}{m}$ submatrices of size m .
- $Y_{n,m}(A)$: Vector of eigenvalues of A , repeated $\binom{n-1}{m-1}$ times.

Proof of Theorem 4.5 (Boundary Cases)

Step 1: Boundary Cases ($m = 1$ and $m = n$) The theorem holds trivially at the extremes:

- **Case** $m = 1$: $Y_{n,1}(A)$ is $\lambda(A)$. $X_1(A)$ is the vector of diagonal entries.

$$Y_{n,1}(A) \succ X_1(A) \iff \lambda(A) \succ \text{diag}(A).$$

This is precisely **Schur's Theorem**.

- **Case** $m = n$:

$$Y_{n,n}(A) = \lambda(A) = X_n(A).$$

Equality implies majorization.

Low Dimensions ($n = 1, 2$): For $n = 1$ (where $m = 1$) and $n = 2$ (where $m = 1, 2$), the result is fully covered by these boundary cases.

Proof of Theorem 4.5 (Induction)

Step 2: Induction ($n \geq 3$) We proceed by induction on the dimension n .

Inductive Hypothesis

Assume that for all $(n-1) \times (n-1)$ Hermitian matrices B , and all $1 \leq m \leq n-1$:

$$Y_{n-1,m}(B) \succ X_m(B).$$

Applying the Lemma: By **Lemma 4.4** (The Inductive Link), if the relation holds for size $n-1$, it implies:

$$Y_{n,m}(A) \succ X_m(A)$$

for all $n \times n$ matrices A and all $1 \leq m \leq n$.

Conclusion: Since the base cases ($n = 1, 2$) are established, the theorem holds for all n by mathematical induction. □

Proof of Spectral Hierarchy (Local Application)

Step 1: Apply Theorem 4.5 Locally

Let M be a specific principal submatrix of A of size $m \times m$. We treat M as a Hermitian matrix in its own right. Apply **Theorem 4.5** to M with a smaller subsize k ($1 \leq k \leq m$).

We have $Y_{m,k}(M) \succ X_k(M)$, which expands to:

$$\underbrace{\lambda(M) \cup \cdots \cup \lambda(M)}_{\binom{m-1}{k-1} \text{ times}} \succ X_k(M), \quad (7)$$

where $X_k(M)$ is the vector of eigenvalues of all $k \times k$ submatrices contained within M .

Proof of Spectral Hierarchy (Aggregation)

Step 2: Take the Union

We take the union of the local relationship (7) over **all** $\binom{n}{m}$ possible principal submatrices M of A .

Analyzing the Left-Hand Side (LHS):

- The union collects the vectors $\lambda(M)$ for every submatrix M .
- By definition, the collection of all $\lambda(M)$ is exactly the vector $X_m(A)$.
- Since each local term repeated $\lambda(M)$ exactly $\binom{m-1}{k-1}$ times, the aggregate LHS is:

$$\underbrace{X_m(A) \cup \dots \cup X_m(A)}_{\binom{m-1}{k-1} \text{ times}}.$$

Proof of Spectral Hierarchy (Counting Parents)

Analyzing the Right-Hand Side (RHS): The RHS collects every $k \times k$ submatrix found inside every $m \times m$ parent M .

The Counting Argument: Let Q be a specific $k \times k$ principal submatrix of A . How many times does its spectrum appear in the union?

- Q appears whenever we select a parent M that contains Q .
- To form such a parent M , we must include the k indices of Q .
- We must choose the remaining $m - k$ indices from the available $n - k$ indices of A .

Total number of parents containing Q :

$$\text{Count} = \binom{n - k}{m - k}.$$

Proof of Spectral Hierarchy (Conclusion)

Step 3: Final Assembly

Since every distinct $k \times k$ submatrix Q is counted exactly $\binom{n-k}{m-k}$ times, the aggregate RHS is equivalent to:

$$\underbrace{X_k(A) \cup \dots \cup X_k(A)}_{\binom{n-k}{m-k} \text{ times}}.$$

Combining the LHS and RHS yields the Theorem:

$$\underbrace{X_m(A) \cup \dots \cup X_m(A)}_{\binom{m-1}{k-1} \text{ times}} \succ \underbrace{X_k(A) \cup \dots \cup X_k(A)}_{\binom{n-k}{m-k} \text{ times}}.$$



Dessert

A Concrete Numerical Tasting

The Ingredients: A Concrete 4×4 Matrix

Instead of assuming values, let's construct a block-diagonal matrix A :

$$A = \left(\begin{array}{cc|cc} 6 & 2 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ \hline 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

1. The Eigenvalues of A :

Since A is block diagonal, its spectrum is the union of the spectra of the two 2×2 blocks.

- Top Block: Roots of $\lambda^2 - 9\lambda + 14 = 0 \implies \mathbf{7, 2}$.
- Bottom Block: Roots of $\lambda^2 - 4\lambda + 3 = 0 \implies \mathbf{3, 1}$.

$$\lambda(A) = (7, \quad 3, \quad 2, \quad 1).$$

The Submatrices (Calculated)

Deleting a row/column from a block diagonal matrix is easy. It breaks one block into a scalar, leaving the other block intact.

The 4 Principal Submatrices (3×3):

- A_1 (**Delete row 1**): Leaves scalar 3 and the full Bottom Block (3, 1).
 $\implies \mu(A_1) = (3, 3, 1).$
- A_2 (**Delete row 2**): Leaves scalar 6 and the full Bottom Block (3, 1).
 $\implies \mu(A_2) = (6, 3, 1).$
- A_3 (**Delete row 3**): Leaves Top Block (7, 2) and scalar 2.
 $\implies \mu(A_3) = (7, 2, 2).$
- A_4 (**Delete row 4**): Leaves Top Block (7, 2) and scalar 2.
 $\implies \mu(A_4) = (7, 2, 2).$

Verification: Sum of all $\mu = 7 + 23 + 9 = 39.$

$(n-1)\text{tr}(A) = 3 \times (6 + 3 + 2 + 2) = 3 \times 13 = 39. \checkmark$

Course 1: The Power of Aggregation (Theorem 3.5)

To see the improvement over Thompson, we sum the **1st and 2nd largest eigenvalues** of all submatrices.

1. The Actual Sum:

$$\sum_{k=1}^4 (\mu_{k,1} + \mu_{k,2}) = (3 + 3) + (6 + 3) + (7 + 2) + (7 + 2) = \mathbf{33}.$$

2. The New Aggregate Bound (Thm 3.5):

$$\text{UB}_{\text{New}} = (n-1)(\lambda_1 + \lambda_2) + (r-\ell+1)\lambda_3 = 3(7+3) + 2(2) = \mathbf{34}.$$

3. The Naive Thompson Bound (Summing separate bounds):

$$\text{UB}_{\text{Naive}} = [(n-1)\lambda_1 + \lambda_2] + [(n-1)\lambda_2 + \lambda_3] = 24 + 11 = \mathbf{35}.$$

$$\text{Actual}(\mathbf{33}) \leq \text{New}(\mathbf{34}) < \text{Naive}(\mathbf{35})$$

Conclusion: Our aggregate bound is strictly sharper than simply adding up Thompson's inequalities!

Course 2: Spectral Hierarchy (Theorem 4.1)

We check if $Y_{4,3}(A) \succ X_3(A)$.

Vector $Y_{4,3}$ (Parent)

$\lambda(A)$ repeated $\binom{3}{2} = 3$ times.

Sorted:

① 7, 7, 7

② 3, 3, 3

③ 2, 2, 2

④ 1, 1, 1

Sum = 39.

Vector X_3 (Children)

Collection of all submatrix eigenvalues.

Sorted:

① 7, 7, 6 $\leftarrow (20 < 21)$

② 3, 3, 3

③ 2, 2, 2

④ 1, 1, 1

Sum = 39.

Majorization Check: The partial sums of Y are always $\geq$ partial sums of X .

• $k = 1: 7 \geq 7 \checkmark$

• $k = 3: 21 \geq 7 + 7 + 6 = 20 \checkmark$

The hierarchy holds perfectly.

Merci de votre attention!

Thank you for your attention!

References I



R. Bhatia. *Matrix Analysis*, Graduate Texts in Mathematics, vol. 169, Springer, New York, 1997.



J. R. Bunch, C. P. Nielsen, and D. C. Sorensen. Rank-one modification of the symmetric eigenproblem, *Numer. Math.*, **31**, 31–48, (1978).



P. B. Denton, S. J. Parke, T. Tao, and X. Zhang. Eigenvectors from eigenvalues: a survey of a basic identity in linear algebra, *Bull. Amer. Math. Soc.*, **59**, 31–58, (2022).



R. A. Horn and C. R. Johnson. *Matrix Analysis*, 2nd ed., Cambridge University Press, Cambridge, 2013.



C. R. Johnson and H. A. Robinson. Eigenvalue inequalities for principal submatrices, *Linear Algebra Appl.*, **37**, 11–22, (1981).



E.S. Katsoprinakis. Geometry of Polynomials and Majorization Theory, *Complex and harmonic analysis: Proceedings of the International Conference, May 25–27, 2007, Aristotle University of Thessaloniki*, 111–128, (2007).



A.W. Marshall, I. Olkin, and B.C. Arnold. *Inequalities: Theory of Majorization and Its Applications*, 2nd ed., Springer, 2011.



P. Nylen, T.-Y. Tam, and F. Uhlig. On the eigenvalues of principal submatrices of normal, Hermitian and symmetric matrices, *Linear and Multilinear Algebra*, **36**, 69–78, (1993).



R. C. Thompson. Principal submatrices of normal and Hermitian matrices, *Illinois J. Math.*, **10**, 296–308, (1966).



R. C. Thompson and P. McEntegert. Principal submatrices II: The Upper and Lower Quadratic Inequalities, *Linear Algebra Appl.*, **1**, 211–243, (1968).



R. C. Thompson. Principal submatrices III: Linear Inequalities, *J. Res. Natl. Bur. Stand. Sec. B*, **72B**(1), 7–22, (1968).

References II



R. C. Thompson. The eigenvalue spreads of a Hermitian matrix and its principal submatrices, *Linear and Multilinear Algebra*, **32**, 327–333, (1992).



R. C. Thompson and S. Therianos. Inequalities connecting the eigenvalues of a hermitian matrix with the eigenvalues of complementary principal submatrices, *Bull. Austral. Math. Soc.*, **6**, 117–132, (1972).



G. H. Golub. Some modified matrix eigenvalue problems, *SIAM Rev.*, **15**(2), 318–334, (1973).